

Mathematical Introduction to Machine Learning

FINAL PROJECT

Instructor: Lei Wu

Due: See below

Please let me know if you find anything is unclear!

Abstract

This final project is designed to equip students with practical experience in training neural networks, honing their ability to systematically analyze complex learning phenomena, and developing skills in writing clear, well-organized reports. When preparing your final report, consider whether you have effectively strengthened these abilities—if not, there may still be room for improvement. Our grading policy will also reflect this emphasis on practical skills, analytical depth, and clarity of presentation.

1 Project Format

- The project should be completed in groups of up to 3 people. Please register your group in <https://docs.qq.com/sheet/DTWdUQUVEb0NFZ3dS?tab=BB08J2> and indicate which members, if any, are graduate students.
- The submitted report should be a PDF file prepared using the **NeurIPS** LaTeX template (mandatory). The report should be of around **4-7 pages** in the pdf format. Unlimited references and appendices are allowed but we will not grade your report based on them. Please note that the 4-7 page limit is strict.

Please also mention the detailed group information, including group members' names and PKU IDs, in your report. Both Chinese and English are allowed for your report but the **group information should be in Chinese**.

- Please complete the project using Python, preferably using PyTorch.

2 Submission

Please send your report and code to all the TAs:

- Zilin Wang (wangzilin@stu.pku.edu.cn)
- Yuhao Liu (yh-liu21@mails.tsinghua.edu.cn)
- Binghui Li (libinghui@pku.edu.cn)

2.1 Deadlines

- The submission deadline for the report and supplementary materials is **January 10, 2026**.

This deadline is tentative and may be adjusted if necessary.

2.2 Requirements

- You are required to compress the materials into a file named `supplementary.zip` and submit it as supplementary material along with your report. After decompression, the directory structure should be as follows:

```
supplementary/  
  code/  
    README.md  
    your code (*.py, etc.)  
  doc/  
    report.pdf  
    else (*.tex, *.bib, etc.)
```

- In the `README.md` file, you need to specify:
 - How to run your main program (e.g., `python main.py --arg1 xxx --arg2 xxx`) to ensure we can reproduce your results.
 - The primary content and purpose of each `.py` file.
 - The versions of Python/modules used in your program. If non-Anaconda modules are used, include relevant details in `README.md`.
 - Any additional necessary information.

3 Project Tasks (total 60 points)

Reproduce the experiments in the paper: [Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets](#) and beyond. You can start by referring to the official code <https://github.com/openai/grok>¹ and some other github repos.

3.1 The problem

Specifically, this project only needs to explore the learning of [modular addition](#):

$$(x, y) \mapsto (x + y) \bmod p, \quad \text{for } x, y \in \mathbb{Z}_p.$$

Keep in mind that there are two key factors impacting the learnability of this problem.

- The prime number p : Larger p corresponds a harder problem.
- The training data fraction:

$$\alpha = \frac{\text{size of training data}}{\text{size of total data}}.$$

Note that the size of total data is p^2 .

3.2 Tasks

Specifically, you need to at least perform the following subtasks:

- Subtask 1 (**10 points**): Reproduce the left part of Figure 1 of the paper for transformer models using the [AdamW](#) optimizer. Please also examine how the data fraction α impacts the grokking phenomenon.

¹This codebase is outdated and it probably does not work well with current pytorch version

- Subtask 2 (**15 points**): Investigate the grokking phenomenon for other network architectures, such as MLP and LSTM.
- Subtask 3 (**15 points**): Analyze the effects of different optimizers (such as SGD with varying learning rates and batch sizes, RMSprop, momentum-based optimizers, etc.) and regularization techniques (e.g., dropout, weight decay) on the grokking phenomenon. Refer to Figure 2 in the paper.
- Subtask 4 (**10 points**): Consider harder problem

$$(x_1, x_2, \dots, x_K) \mapsto \left(\sum_{k=1}^K x_k \right) \bmod p,$$

where $x_k \in [p]$ for $k = 1, \dots, K$. Investigate the impact of the number of summands: K on the grokking phenomenon.

- Subtask 5 (**10 points**): Provide an explanation of the grokking phenomenon. You can refer to relevant research papers but you need to use your own language and experiments.

To aid your work, we have included three exemplary reports from last year. Please use them for guidance, but **avoid direct replication**.

4 Additional Remarks

- You can choose $p \in \mathbb{N}$ as small as possible to reduce computational cost as long as it does not affect your conclusions.
- It should be stressed that “integers” are not assumed to be known to the models, and you will need to encode them as one-hot vectors. For more details, please refer to the paper and official code.
- You can rent GPUs from online services to facilitate your experiments. Platforms like <https://www.autodl.com/home> and others offer GPU rentals, some of which provide hundreds of GPU hours of free trial usage. You can search for such services using your preferred search engine.